

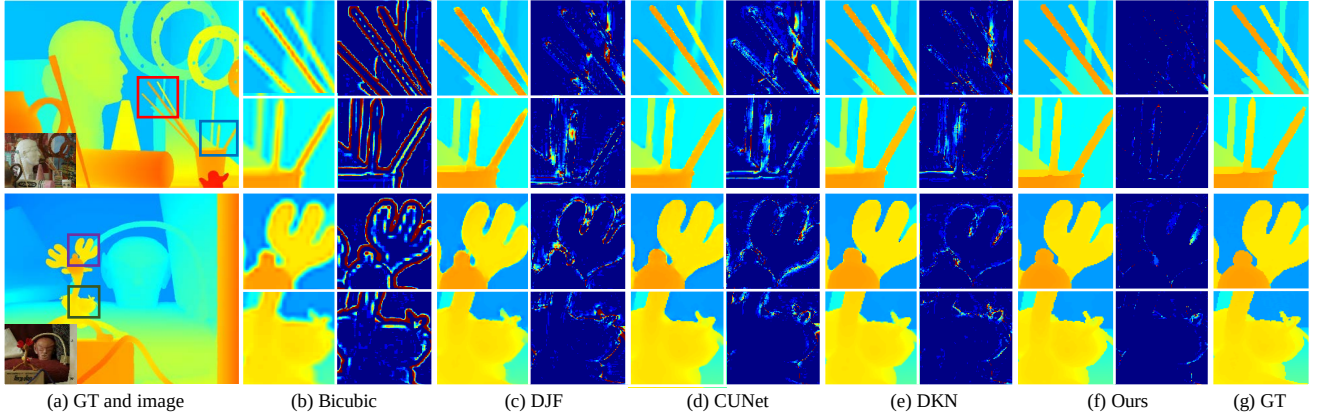


Figure 7: Visual comparisons of Art and Laundry on Middlebury dataset ($\times 8$ case).

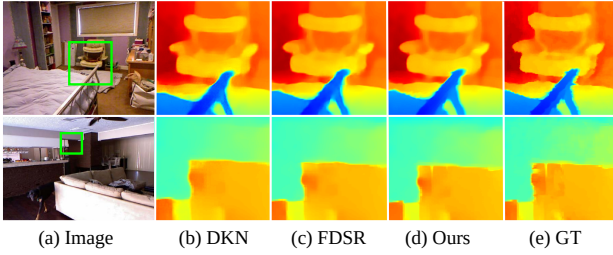


Figure 8: Visual comparisons on NYU-v2 dataset ($\times 8$ case).

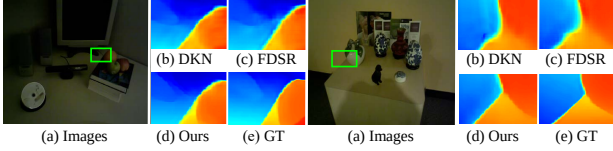


Figure 9: Visual comparisons on Lu dataset ($\times 16$ case).

sistent image features. We can see that the SA module can outperform them by a large margin. From the results of Table 4, it is observed that all the modules proposed in the RSAG framework have made a positive contribution to the ultimate success of our method. To further study the impact of the recurrent steps of SA, we conduct experiments on Middlebury and NYU-v2 datasets by varying the step from 0 (w/o SA) to 4, as illustrated in Fig. 10. It can be found that the method achieves better performance when the recursion steps increase, where 2 recurrent steps obtain the best trade-off between speed and accuracy. It also proves that higher-quality depth information can help obtain a more reliable guidance structure for subsequent depth reconstruction.

Conclusion

In this paper, we proposed a novel recurrent structure attention guided (RSAG) framework for depth super-resolution. In our framework, a deep contrastive network with multi-scale filters (DCN) block was designed to adaptively de-

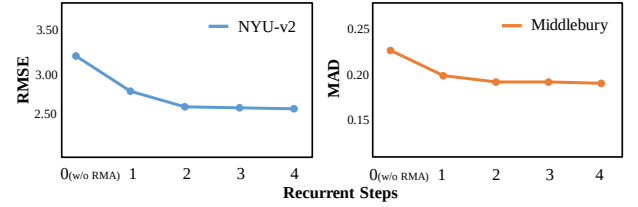


Figure 10: Ablation studies of SA with different recursive steps on Middlebury and NYU-v2 datasets ($\times 8$ case).

compose the high-quality HF and LF components by using contrastive networks from large kernels to small ones. In addition, by leveraging the latest depth output and high-resolution image as guidance, we introduced recurrent structure attention (SA) block, instead of the trivial feature concatenation, to select consistent and clear image features for subsequent cross-modality fusion. Furthermore, we presented the HF&LF feature fusion block to refine the blurred edges of the LF component. Extensive experiments on various benchmark datasets demonstrated the superiority and effectiveness of the proposed framework.

Acknowledgement

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